

Mathematical foundation of Error detection & Correction Syllabus

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Date: _____

Unit-I Communication channels & maximum Likelihood Decoding

- Communication channels
- Types of errors in digital communication
- Maximum likelihood decoding

Unit-II Distance based error correction

- Hamming distance & its significance
- Nearest neighbour (minimum distance) decoding
- Distance of a code & its role in error detection

Unit-III Finite field & Polynomial rings

- Basics of fields & Polynomial rings
- Structure & properties of finite fields
- Minimal polynomials & their applications

Unit-IV Linear codes & Decoding methods

- Vector space over finite fields
- Generator matrix & parity check matrix
- Encoding & decoding techniques
- Cosets, Syndrome decoding & nearest neighbour decoding for linear codes

Books

- 1) Coding theory: A first course by SANLING
- 2) A first course in coding mathematical methods

Set : A collection of well defined objects (elements, members)
notation : $\{ \}$

- write a set of alphabets capital letters.
 $A = \{ A, B, C, D, \dots, X, Y, Z \}$
 $B = \{ 1, 2, 3, 4, 5, \dots, 9, 10 \}$ - Tabular form
 $B = \{ x \in \mathbb{N} \mid x \leq 11 \}$ - set builder method.

Is $6 \in B \rightarrow$ member of B / belongs to B .
 $11 \notin B \rightarrow$ does not belongs to B .

$$C = \{ 3, 5, 7, 9, 10 \}$$

$C \subseteq B$ $C \subset B$
 $\hookrightarrow$ subset $\hookrightarrow$ proper set

Universal set (U):
Set of all possible element under consideration in a particular context.

In coding theory For 3-bit message the universal set is
 $U = \{ 000, 001, 010, 100, 110, 101, 011, 111 \}$

* Power set $P(A)$
The set of all possible subsets of set A including the empty set & the set itself.

Note: If set A contains n elements then total number of subset = 2^n

$\mathbb{Z}$ - Set of Integers

eg. $A = \{ 2, 4 \} \Rightarrow 2^n = 4$

Set Operation

1) Union ($\cup$)

The set of all elements that are in A set or in set B or in both
Notation : $A \cup B = \{ x \mid x \in A \text{ or } x \in B \}$

$$\begin{aligned} A &= \{ 1, 2, 3 \} \\ B &= \{ a, b, c \} \\ A \cup B &= \{ 1, 2, 3, a, b, c \} \\ C &= \{ 3, 4, 5 \} \\ A \cup C &= \{ 1, 2, 3, 3, 4, 5 \} \end{aligned}$$

2) Intersection ($\cap$)

The set of all possible elements that are common to both set A & B .

Notation :
 $A \cap B = \{ x \mid x \in A \text{ and } x \in B \}$

eg. $A = \{ 1, 2, 3 \}$
 $B = \{ a, b, c \}$
 $A \cap B = \emptyset$

$$\begin{aligned} C &= \{ 2, 3, 10 \} \\ A \cap B &= \{ 2, 3 \} \end{aligned}$$

$$\begin{aligned} X &= \{ a, e, i, o, u \} \\ Y &= \{ A, k, s, h, a, d, a \} \\ X \cap Y &= \{ a \} \end{aligned}$$

3) Difference (\) (-)

The set of all possible elements that are in A set but not in B.

• Notation :

$$A - B = A \setminus B = \{x \mid x \in A \text{ but } x \notin B\}$$

e.g. $A = \{1, 2, 3\}$

$$B = \{a, b, c\}$$

$$A - B = \{1, 2, 3\}$$

$$C = \{1, 3, 4\}$$

$$A - C = \{2\}$$

4) Complement (A^c or A')

The set of all elements in the universal set U that are not in set A.

$$A' = U - A$$

e.g. $A = \text{set of vowels in english alphabets.}$

$$A' = \{b, c, d, f, g, h, j, k, l, m, n, p, q, r, s, t, v, w, x, y, z\}$$

Cartesian product

The cartesian product $A \times B$
 $= \{(a, b) \mid a \in A, b \in B\}$

e.g. $A = \{1, 2\}$

$$B = \{0, 1\}$$

$$A \times B = \{(1, 0), (1, 1), (2, 0), (2, 1)\}$$

$$B \times A = \{(0, 1), (0, 2), (1, 1), (1, 2)\}$$

Communication channels & maximum Likelihood Principles / Decoding

Unit-I

Introduction

- In coding theory we study to detect even correct errors. usually coding is defined as source coding & channel coding.

- Source coding involves changing the message source to a suitable code to be transmitted through the channel.

- An example of source coding is the ASCII Code which converts each character to a byte of 8 bits.

Ex - Consider the source coding of 4 messages as follows.

$$M_1 = 00, M_2 = 01, M_3 = 10, M_4 = 11$$

- Suppose, the message m_1 which is encoded as 00 is transmitted over noisy channel.

- The message may be come distorted & may be receive as 01.

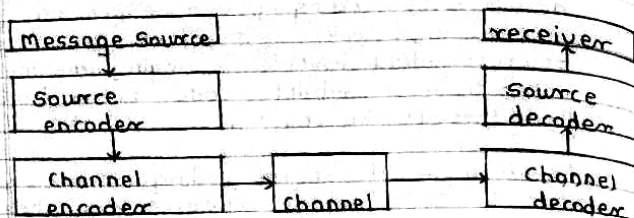
- The receiver may not realize that the message was corrupted.

Therefore, this communication fails.

Therefore, there is new concept channel coding is to encode the message again after the source coding by introducing some form of new encoders, so that errors

can be detected or even corrected.

Therefore, new transmission channel is as follows.



By using above we perform channel encoding by introducing one more bit as follows.

$M_1 - 00 - 000$

$M_2 - 01 - 011$

$M_3 - 10 - 101$

$M_4 - 11 - 110$

As we see the message M_1 is encoded as 000 after the source & channel encoding, is transmitted & if there are one error introduced.

Then the received message is of the form

100

010

001

In this way we can detect the error as none of 100, 010 or 001 is among our encoded messages.

In this way by continuing this process we have to add ^{fixed} numbers of bits in encoded message.

Assume that source coding has already done by sources & the information consists of bit strings of fixed length k .

encoding is carried out by taking bit strings & repeating it $2r+1$ times where $r \geq 0$ is a fixed integer.

for ex- 01 is encoded as,
since, length of bit is 2,
 $\therefore k=2$
& fixed $r=2$
 $\therefore 2r+1 = 4+1 = 5$
 $\therefore 01 - 01010101$

Goal # The goal of channel coding

To construct encoders & decoders in a such way as to effect

- 1) Fast encoding of message
- 2) Easy transmission of encoded messages
- 3) Fast decoding of received messages.
- 4) maximum transfer of information per unit time.
- 5) maximal detection or correction capacity

Communication channels

Basic definitions -

Let $A = \{a_1, a_2, \dots, a_q\}$ be a set of size q , which we refer to as a code alphabet & whose elements are called code symbols.

A q -ary word of length n over A is a sequence $w = w_1 w_2 \dots w_n$ with each $w_i \in A$, $\forall i$. equivalently w is the vector $(w_1, w_2, \dots, w_n)$.

A q -ary block code of length n over A is a non-empty set C of q -ary words having the same length n .

An element of C is called a code word in C .

The number of code words in C is called the size of C , denoted by $|C|$.

The information rate of a code C of length n is defined to be $(\log_q |C|) / n$.

A code of length n & size M is called an (n, M) code.

Ex- $A = \{0, 1\}$ 0 & 1 are code symbols
 $w = \{0, 1\}$
 $C = \{00, 01, 10, 11\}$

Code words.

$$|C| = 4$$

$$n = 2$$

$$q = 2$$

- length of code words.

- Element of A .

$$\begin{aligned} \text{Information rate} &= (\log_q |C|) / n \\ &= (\log_2 4) / 2 \\ &= \frac{\log_2(4)}{2} \\ &= \frac{\log_2(2^2)}{2} \\ &= \frac{2 \log_2(2)}{2} \\ &= 1 \end{aligned}$$

Definition

A communication channel consist of finite channel alphabet $A = \{a_1, a_2, \dots, a_q\}$ as well as a set of forward channel probabilities $P(a_j \text{ recieved} | a_i \text{ sent})$ satisfying

$$P(a_j \text{ recieved} | a_i \text{ sent}) = 1 \quad \text{for all } i$$

where $P(a_j \text{ recieved} | a_i \text{ sent})$ is the conditional probability that a_j is recieved given that a_i is sent.

Definition :

A communication channel is said to be memoryless if the outcome of any one transmission is independent of the outcome of the previous transmission.

i.e. if $C = C_1 C_2 \dots C_n$ &
 $X = X_1 X_2 \dots X_n$ are words of length n .

$$\text{Then } P(X \text{ recieved} / C \text{ sent}) = \prod_{i=1}^n P(X_i \text{ recieved} / C_i \text{ sent})$$

Definition :

A q -ary symmetric channel is a memoryless channel which has a channel alphabet of size q such that

- 1) Each symbol transmitted has the same probability $P(< 1/2)$ of being recieved in error.
- 2) If a symbol is recieved in error then each of the $q-1$ possible errors is equally likely.

Binary Symmetric Channel (BSC)

- It is type of memoryless channel which has channel alphabet $\{0, 1\}$
- It is noisy communication channel where sometimes bits get flipped according to channel probabilities.
- Channel probabilities are,

$$P(1 \text{ recieved} / 0 \text{ sent}) = P(0 \text{ recieved} / 1 \text{ sent}) = P$$

$$P(0 \text{ recieved} / 0 \text{ sent}) = P(1 \text{ recieved} / 1 \text{ sent}) = 1 - P$$

Thus, the probability of a bit error in a BSC is P .

This is called the cross over probability of the BSC.

Ex - Suppose that codewords from the code $\{000, 111\}$ are sent over BSC with cross over probability $P = 0.05$. Suppose that the word 110 is recieved. Then find right code word sent by computing the forward channel probabilities.

Solⁿ: $P(110 \text{ recieved} / 000 \text{ sent})$
 $= P(1 \text{ recieved} / 0 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \times P(0 \text{ recieved} / 0 \text{ sent})$
 $= P \times P \times (1 - P)$
 $= 0.05 \times 0.05 \times (1 - 0.05)$
 $= 0.05 \times 0.05 \times 0.95$
 $= 0.002375$

$$\begin{aligned} P(110 \text{ recieved} / 111 \text{ sent}) &= P(1 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 1 \text{ sent}) \times P(0 \text{ recieved} / 1 \text{ sent}) \\ &= (1 - P) \times (1 - P) \times P \\ &= (1 - 0.05) \times (1 - 0.05) \times 0.05 \\ &= 0.95 \times 0.95 \times 0.05 \\ &= 0.045125 \end{aligned}$$

Since, the code word 111 has 0.045125 probability to give 110 &
 $\therefore$ correct codeword is 111.

Definition :

A communication channel is said to be memoryless if the outcome of any one transmission is independent of the outcome of the previous transmission.

i.e. if $C = C_1 C_2 \dots C_n$ &
 $X = x_1 x_2 \dots x_n$ are words of length n .

$$\text{Then } P(X \text{ received} / C \text{ sent}) = \prod_{i=1}^n P(x_i \text{ received} / C_i \text{ sent})$$

Definition :

A q -ary symmetric channel is a memoryless channel which has a channel alphabet of size q such that

- 1) Each symbol transmitted has the same probability $P(< 1/2)$ of being received in error.
- 2) If a symbol is received in error then each of the $q-1$ possible errors is equally likely.

Binary Symmetric Channel (BSC)

- It is type of memoryless channel which has channel alphabet $\{0, 1\}$
- It is noisy communication channel where sometimes bits get flipped according to channel probabilities.
- Channel probabilities are,

$$P(1 \text{ received} / 0 \text{ sent}) = P(0 \text{ received} / 1 \text{ sent}) = p$$

$$P(0 \text{ received} / 0 \text{ sent}) = P(1 \text{ received} / 1 \text{ sent}) = 1 - p$$

Thus, the probability of a bit error in a BSC is p .

This is called the cross over probability of the BSC.

Ex - Suppose that codewords from the code $\{000, 111\}$ are sent over BSC with cross over probability $p = 0.05$. Suppose that the word 110 is received. Then find right code word sent by computing the forward channel probabilities.

$$\begin{aligned} \text{Soln: } P(110 \text{ received} / 000 \text{ sent}) &= P(1 \text{ received} / 0 \text{ sent}) \times P(1 \text{ received} / 0 \text{ sent}) \\ &\quad \times P(0 \text{ received} / 0 \text{ sent}) \\ &= p \times p \times (1 - p) \\ &= 0.05 \times 0.05 \times (1 - 0.05) \\ &= 0.05 \times 0.05 \times 0.95 \\ &= 0.002375 \end{aligned}$$

$$\begin{aligned} P(110 \text{ received} / 111 \text{ sent}) &= P(1 \text{ received} / 1 \text{ sent}) \times P(1 \text{ received} / 1 \text{ sent}) \\ &\quad \times P(0 \text{ received} / 1 \text{ sent}) \end{aligned}$$

$$\begin{aligned} &= (1 - p) \times (1 - p) \times p \\ &= (1 - 0.05) \times (1 - 0.05) \times 0.05 \\ &= 0.95 \times 0.95 \times 0.05 \\ &= 0.045125 \end{aligned}$$

Since, the code word 111 has 0.045125 probability to give 110 & correct code word is 111.

In above example, we find forward probabilities for each code word & compare them. The greater probability shows which one is correct. This rule is called as Maximum likelihood decoding rule.

IMP # Maximum Likelihood decoding rule

Suppose that codewords from a code C are being sent over a communication channel. If a word x is received we can compute the forward channel probabilities $P(x \text{ received} / c \text{ sent})$ for all the code words $c \in C$.

The MLD rule will conclude that C_x is most likely codeword transmitted if C_x maximizes the forward channel probabilities

$$\text{i.e. } P(x \text{ received} / C_x \text{ sent}) = \max_{c \in C} P(x \text{ received} / c \text{ sent})$$

$$\forall c \in C$$

• Types of MLD

1) Complete maximum likelihood decoding (CMLD):

If a word x is received, find the most likely code word transmitted. If there are more than 2 such codeword select anyone of them.

2) Incomplete Maximum likelihood decoding (IMLD)

If a word x is received find the most likely codeword transmitted.

If there are more than one such codeword request retransmission.

Q. Suppose that code words from binary code $\{000, 100, 111\}$ are being sent over BSC with crossover probability $p = 0.03$. Use the maximum likelihood decoding rule to decode the following received word.
a) 010 b) 011 c) 001

$$\text{a) } P(010 \text{ received} / 000 \text{ sent}) = P(0 \text{ received} / 0 \text{ sent}) \times P(1 \text{ received} / 0 \text{ sent}) \times P(0 \text{ received} / 0 \text{ sent})$$

$$= (1-p) \times p \times (1-p) \\ = (1-0.03) \times 0.03 \times (1-0.03) \\ = 0.028227$$

$$\text{b) } P(010 \text{ received} / 100 \text{ sent}) = P(0 \text{ received} / 1 \text{ sent}) \times P(1 \text{ received} / 0 \text{ sent}) \times P(0 \text{ received} / 0 \text{ sent}) \\ = (1-p) \times (1-p) \times (1-p) \\ = (1-0.03) \times (1-0.03) \times (1-0.03)$$

$$= p \times p \times p \\ = 0.03 \times 0.03 \times 0.03 \\ = 0.000813$$

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$$\begin{aligned} \text{iii)} & P(010 \text{ recieved} / \text{III sent}) \\ &= P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 1 \text{ sent}) \\ &\quad \times P(0 \text{ recieved} / 1 \text{ sent}) \\ &= P \times (1-P) \times P \\ &= 0.000873 \end{aligned}$$

By MLD rule 000 is correct code word which is transmitted.

b) i) $P(011 \text{ recieved} / 000 \text{ sent})$

$$\begin{aligned} &= P(0 \text{ recieved} / 0 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \times \\ &\quad \times P(1 \text{ recieved} / 0 \text{ sent}) \\ &= (1-P) \times P \times P \\ &= (1-0.03) \times 0.03 \times 0.03 \\ &= 0.97 \times 0.03 \times 0.03 \\ &= 0.000873 \end{aligned}$$

ii) $P(011 \text{ recieved} / 100 \text{ sent})$

$$\begin{aligned} &= P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \\ &\quad \times P(1 \text{ recieved} / 0 \text{ sent}) \\ &= P \times P \times P \\ &= 0.03 \times 0.03 \times 0.03 \\ &= 0.000027 \end{aligned}$$

iii) $P(011 \text{ recieved} / \text{III sent})$

$$\begin{aligned} &= P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 1 \text{ sent}) \\ &\quad \times P(1 \text{ recieved} / 1 \text{ sent}) \\ &= P \times (1-P) \times (1-P) \\ &= 0.03 \times (1-0.03) \times (1-0.03) \\ &= 0.000027 \\ &\quad 0.03 \times (0.97) \times 0.97 \\ &= 0.028227 \end{aligned}$$

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By MLD rule III is correct code word which is transmitted.

c) i) $P(001 \text{ recieved} / 000 \text{ sent})$

$$\begin{aligned} &= P(0 \text{ recieved} / 0 \text{ sent}) \times P(0 \text{ recieved} / 0 \text{ sent}) \\ &\quad \times P(1 \text{ recieved} / 0 \text{ sent}) \\ &= (1-P) \times (1-P) \times P \\ &= (1-0.03) \times (1-0.03) \times 0.03 \\ &= 0.97 \times 0.97 \times 0.03 \\ &= 0.028227 \end{aligned}$$

ii) $P(001 \text{ recieved} / 100 \text{ sent})$

$$\begin{aligned} &= P(0 \text{ recieved} / 1 \text{ sent}) \times P(0 \text{ recieved} / 0 \text{ sent}) \\ &\quad \times P(1 \text{ recieved} / 0 \text{ sent}) \\ &= P \times (1-P) \times P \\ &= 0.03 \times (1-0.03) \times 0.03 \\ &= 0.03 \times 0.97 \times 0.03 \\ &= 0.000873 \end{aligned}$$

iii) $P(001 \text{ recieved} / \text{III sent})$

$$\begin{aligned} &= P(0 \text{ recieved} / 1 \text{ sent}) \times P(0 \text{ recieved} / 1 \text{ sent}) \\ &\quad \times P(1 \text{ recieved} / 1 \text{ sent}) \\ &= P \times P \times (1-P) \\ &= 0.03 \times 0.03 \times 0.97 \\ &= 0.000873 \end{aligned}$$

By MLD rule 000 is correct code word which is transmitted.

$$\begin{aligned} \text{ii)} P(010 \text{ recieved} / 111 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 1 \text{ sent}) \\ \times P(0 \text{ recieved} / 1 \text{ sent}) \\ = P \times (1-P) \times P \\ = 0.00873 \end{aligned}$$

By MLD rule 000 is correct code word which is transmitted.

$$\begin{aligned} \text{b) i) } P(011 \text{ recieved} / 000 \text{ sent}) \\ = P(0 \text{ recieved} / 0 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \times \\ P(1 \text{ recieved} / 0 \text{ sent}) \\ = (1-P) \times P \times P \\ = (1-0.03) \times 0.03 \times 0.03 \\ = 0.97 \times 0.03 \times 0.03 \\ = 0.000873 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(011 \text{ recieved} / 100 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \\ \times P(1 \text{ recieved} / 0 \text{ sent}) \\ = P \times P \times P \\ = 0.03 \times 0.03 \times 0.03 \\ = 0.00027 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(011 \text{ recieved} / 111 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 1 \text{ sent}) \\ \times P(1 \text{ recieved} / 1 \text{ sent}) \\ = P \times (1-P) \times (1-P) \\ = 0.03 \times (1-0.03) \times (1-0.03) \\ = 0.00027 \\ 0.03 \times (0.97) \times 0.97 \\ = 0.028227 \end{aligned}$$

By MLD rule 111 is correct code word which is transmitted.

$$\begin{aligned} \text{c) i) } P(001 \text{ recieved} / 000 \text{ sent}) \\ = P(0 \text{ recieved} / 0 \text{ sent}) \times P(0 \text{ recieved} / 0 \text{ sent}) \\ \times P(1 \text{ recieved} / 0 \text{ sent}) \\ = (1-P) \times (1-P) \times P \\ = (1-0.03) \times (1-0.03) \times 0.03 \\ = 0.97 \times 0.97 \times 0.03 \\ = 0.028227 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(001 \text{ recieved} / 100 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(0 \text{ recieved} / 0 \text{ sent}) \\ \times P(1 \text{ recieved} / 0 \text{ sent}) \\ = P \times (1-P) \times P \\ = 0.03 \times (1-0.03) \times 0.03 \\ = 0.03 \times 0.97 \times 0.03 \\ = 0.000873 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(001 \text{ recieved} / 111 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(0 \text{ recieved} / 1 \text{ sent}) \\ \times P(1 \text{ recieved} / 1 \text{ sent}) \\ = P \times P \times (1-P) \\ = 0.03 \times 0.03 \times 0.97 \\ = 0.000873 \end{aligned}$$

By MLD rule 000 is correct codeword which is transmitted.

Q. Consider memory less binary channel with channel probabilities.

$$P(0 \text{ recieved} / 0 \text{ sent}) = 0.7$$

$$P(1 \text{ recieved} / 1 \text{ sent}) = 0.8$$

If codewords from the binary code {000, 100, 111} are being sent over this channel, news maximum likelihood decoding rule to decode the following recieved word

a) 010 b) 011 c) 001

→ Given

$$P(0 \text{ recieved} / 0 \text{ sent}) = 0.7$$

$$\therefore P(0 \text{ recieved} / 1 \text{ sent}) = 1 - 0.7 = 0.3$$

$$P(1 \text{ recieved} / 1 \text{ sent}) = 0.8$$

$$P(1 \text{ recieved} / 0 \text{ sent}) = 1 - 0.8 = 0.2$$

$$\begin{aligned} \text{a) i) } P(010 \text{ recieved} / 000 \text{ sent}) \\ = P(0 \text{ recieved} / 0 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \\ \times P(0 \text{ recieved} / 0 \text{ sent}) \\ = 0.7 \times 0.2 \times 0.7 \\ = 0.098 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(010 \text{ recieved} / 100 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \\ \times P(0 \text{ recieved} / 0 \text{ sent}) \\ = 0.3 \times 0.2 \times 0.7 \\ = 0.042 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(010 \text{ recieved} / 111 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 1 \text{ sent}) \\ \times P(0 \text{ recieved} / 1 \text{ sent}) \\ = 0.3 \times 0.8 \times 0.3 \\ = 0.072 \end{aligned}$$

By MLD rule 000 correct codeword which is transmitted

$$\begin{aligned} \text{b) i) } P(011 \text{ recieved} / 000 \text{ sent}) \\ = P(0 \text{ recieved} / 0 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \\ \times P(1 \text{ recieved} / 0 \text{ sent}) \\ = 0.7 \times 0.2 \times 0.2 \\ = 0.028 \end{aligned}$$

$$\begin{aligned} \text{ii) } P(011 \text{ recieved} / 100 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 0 \text{ sent}) \\ \times P(1 \text{ recieved} / 0 \text{ sent}) \\ = 0.3 \times 0.2 \times 0.2 \\ = 0.012 \end{aligned}$$

$$\begin{aligned} \text{iii) } P(011 \text{ recieved} / 111 \text{ sent}) \\ = P(0 \text{ recieved} / 1 \text{ sent}) \times P(1 \text{ recieved} / 1 \text{ sent}) \\ \times P(1 \text{ recieved} / 1 \text{ sent}) \\ = 0.3 \times 0.8 \times 0.8 \\ = 0.192 \end{aligned}$$

By MLD rule 111 is correct codeword for which is transmitted

$$\begin{aligned} \text{c) i) } P(001 \text{ recieved} / 000 \text{ sent}) \\ = P(0 \text{ recieved} / 0 \text{ sent}) \times P(0 \text{ recieved} / 0 \text{ sent}) \\ \times P(1 \text{ recieved} / 0 \text{ sent}) \\ = 0.7 \times 0.7 \times 0.2 \\ = 0.098 \end{aligned}$$

$$\begin{aligned}
 & \text{ii) } P(001 \text{ recieved} / 100 \text{ sent}) \\
 &= P(0 \text{ recieved} / 1 \text{ sent}) \times P(0 \text{ recieved} / 0 \text{ sent}) \\
 &\quad \times P(1 \text{ recieved} / 0 \text{ sent}) \\
 &= 0.3 \times 0.7 \times 0.2 \\
 &= 0.042
 \end{aligned}$$

$$\begin{aligned}
 & \text{iii) } P(001 \text{ recieved} / 111 \text{ sent}) \\
 &= P(0 \text{ recieved} / 1 \text{ sent}) \times P(0 \text{ recieved} / 1 \text{ sent}) \\
 &\quad \times P(1 \text{ recieved} / 1 \text{ sent}) \\
 &= 0.3 \times 0.3 \times 0.8 \\
 &= 0.072
 \end{aligned}$$

By MLD rule 000 is correct codeword which is transmitted.

• Types of Error in digital communication

- 1) Bit Error
 - In this error one or more bit changes.
 - Single bit Error is most common in this type.

Ex - Sent 1 but recieved 0

- 2) Burst Error
 - In this type of error several bits change together.

Ex - sent - 1011 but recieved - 1000

- 3) Packet Loss Error
 - In this type of error a whole message or part of message is missing.

Ex - { m_1, m_2, m_3 } is sent but { m_1, m_3 } recieved.

- 4) Synchronisation Error
 - In this type of error receiver reads message wrongly

Ex -

- 5) jitter
 - In this type of error data arrives unevenly

ex - Delayed video or sound.

Unit-II Distance based Error detection

Hamming distance

Let x & y keywords of length n over an alphabet A .

The having distance from x to y is denoted by $d(x, y)$ is defined to be the no. of places at which x & y is different.

If $x = x_1 \dots x_n$

$y = y_1 \dots y_n$

then $d(x, y) = d(x_1, y_1) + \dots + d(x_n, y_n)$

Where, x_i & y_i are words of length 1

$$d(x_i, y_i) = \begin{cases} 1 & \text{if } x_i \neq y_i \\ 0 & \text{if } x_i = y_i \end{cases}$$

Ex: Let $A = \{0, 1\}$ $x = 01010$ $y = 01101$ $z = 11101$
then find $d(x, y)$, $d(y, z)$, $d(x, z)$

Soln: $x = 01010$
 $y = 01101$
 $z = 11101$

$$d(x, y) = 0 + 0 + 1 + 1 + 1 = 3$$

$$d(y, z) = 1 + 0 + 0 + 0 + 0 = 1$$

$$d(x, z) = 1 + 0 + 1 + 1 + 1 = 4$$

Q.2 $A = \{0, 1, 2, 3, 4\}$ $x = 1234$, $y = 1423$, $z = 3214$
then find $d(x, y)$, $d(y, z)$, $d(x, z)$

Soln: $x = 1234$ $d(x, y) = 0 + 1 + 1 + 1 = 3$
 $y = 1423$ $d(y, z) = 1 + 1 + 1 + 1 = 4$
 $z = 3214$ $d(x, z) = 1 + 0 + 1 + 0 = 2$

Theorem :

Statement: Let x, y, z be words of length n over A , then we have

i) $0 \leq d(x, y) \leq n$

ii) $d(x, y) = 0$ iff $x = y$

iii) $d(x, y) = d(y, x)$

iv) $d(x, z) \leq d(x, y) + d(y, z)$ (Triangle inequality)

Proof: i) $0 \leq d(x, y) \leq n$

If $x = x_1 \dots x_n$ &

$y = y_1 \dots y_n$

Then we have

$$d(x, y) = d(x_1, y_1) + \dots + d(x_n, y_n)$$

$$d(x_i, y_i) = \begin{cases} 1 & , x_i \neq y_i \\ 0 & , x_i = y_i \end{cases}$$

$$0 \leq d(x_i, y_i) \leq 1$$

$$\therefore 0 \leq d(x_1, y_1) + d(x_2, y_2) + \dots + d(x_n, y_n) \leq 1 + 1 + \dots + 1 \text{ (n-times)}$$

$$\Rightarrow 0 \leq d(x, y) \leq n$$

ii) $d(x, y) = 0$ iff $x = y$

First we assume that -

$$d(x, y) = 0$$

Then we have to show that $x = y$

$$d(x, y) = 0$$

$$\therefore d(x_1, y_1) + d(x_2, y_2) + \dots + d(x_n, y_n) = 0$$

$$\Rightarrow d(x_1, y_1) = d(x_2, y_2) = \dots = d(x_n, y_n) = 0$$

$$\Rightarrow x_1 = y_1, x_2 = y_2, \dots, x_n = y_n$$

$$\Rightarrow x_1 x_2 \dots x_n = y_1 y_2 \dots y_n$$

$$\Rightarrow x = y$$

& vice versa

$$\begin{aligned} \text{iii)} \quad d(x, y) &= d(y, x) \\ d(x, y) &= d(x_1, y_1) + d(x_2, y_2) + \dots + d(x_n, y_n) \\ d(x, y) &= d(x_1, y_1) = d(y_1, x_1) + d(y_2, x_2) \\ &\quad + \dots + d(y_n, x_n) \\ d(x, y) &= d(x_i, y_i) = d(y_i, x_i) = d(y, x) \end{aligned}$$

$$\text{iv)} \quad d(x, z) \leq d(x, y) + d(y, z)$$

a) If $x = z$ then $d(x, z) = 0$
 $\therefore d(x, z) \leq d(x, y) + d(y, z)$

b) $x \neq z, y \neq z, x \neq y$
 $\therefore d(x, z) = 1, d(y, z) = 1, d(x, y) = 1$
 $\therefore d(x, z) \leq d(x, y) + d(y, z)$
 $1 \leq 1 + 1$
 $1 \leq 2$

To show in general

c) $d(x, z) \leq d(x, y) + d(y, z)$
 $\therefore d(x_1, z_1) + \dots + d(x_n, z_n) \leq d(x_1, y_1) + \dots + d(x_n, y_n) + d(y_1, z_1) + \dots + d(y_n, z_n)$
 $\therefore d(x_i, z_i) \leq d(x_i, y_i) + d(y_i, z_i)$
 $\therefore d(x, z) \leq d(x, y) + d(y, z)$

Q.1) Verify properties of hamming distance
 where $x = 10101, y = 10101, z = 11011$

Soln: $x = 10101$
 $y = 10101$
 $z = 11011$

$$\begin{aligned} d(x, y) &= 0 + 0 + 0 + 0 + 0 = 0 \\ d(y, z) &= 0 + 1 + 1 + 1 + 0 = 3 \\ d(x, z) &= 0 + 1 + 1 + 1 + 0 = 3 \end{aligned}$$

1) $0 \leq d(x, y) \leq n$ $n = \text{Length}$
 $\therefore d(x, y) = 0 \leq 5$

2) $d(x, y) = 0$ iff $x = y$
 Where, $x = y = 10101$
 $\therefore d(x, y) = 0$ and vice versa

3) $d(x, y) = d(y, x)$
 we get $d(x, y) = d(y, x) = 0$

4) $d(x, z) \leq d(x, y) + d(y, z)$
 $3 \leq 0 + 3$
 $\therefore 3 \leq 3$
 $= 3 = 3$ is verified

2) Verify properties of hamming distance
 where

$$x = 1011101$$

$$y = 1001001$$

$$z = 1110001$$

Soln: $d(x, y) = 0 + 0 + 1 + 0 + 1 + 0 + 0 = 2$
 $d(y, z) = 0 + 1 + 1 + 1 + 0 + 0 + 0 = 3$
 $d(x, z) = 0 + 1 + 0 + 1 + 1 + 0 + 0 = 3$

$$1) 0 \leq d(x, y) \leq n$$

$$\therefore d(x, y) = 2 < 7$$

$$2) d(x, y) = 0 \text{ iff } x = y$$

here, $x = 1011101$, $y = 1001001$

$$\therefore x \neq y$$

$$\therefore d(x, y) \neq 0$$

$$3) d(x, y) = d(y, x)$$

we get $d(x, y) = d(y, x) = 2$

$$4) d(x, z) \leq d(x, y) + d(y, z)$$

$$3 \leq 2 + 3$$

$$3 < 5 \text{ is verified}$$

Suppose codewords from the binary

Nearest Neighbour :-

Suppose that code words from a code C are being sent over a communication channel.

If a word x is recieved, the nearest Neighbour Decoding rule will decode x to C_x if distance of $d(x, C_x)$ is minimal among the all codewords in C is

$$d(x, C_x) = \min d(x, C)$$

Q. Suppose codewords from the binary are $C = \{0000, 0011, 1000, 1100, 0001, 1001\}$ are sent over BSC, decode $x = 0111$ by nearest neighbour decoding

$$d(x, 0000) = d(0111, 0000) = 3$$

$$d(x, 0011) = d(0111, 0011) = 1$$

$$d(x, 1000) = d(0111, 1000) = 4$$

$$d(x, 1100) = d(0111, 1100) = 3$$

$$d(x, 0001) = d(0111, 0001) = 2$$

$$d(x, 1001) = d(0111, 1001) = 3$$

By nearest neighbour decoding rule minimum distance is 1 for $C_x = 0011$

$\therefore$ decoded codeword is 0011

A. For the Binary code $C = \{01101, 00011, 10110, 11000\}$ use the nearest neighbour decoding rule to decode the following recieved words.

a) 00000

$\rightarrow C = \{01101, 00011, 10110, 11000\}$

$a = 00000$

$$d(a, 01101) = d(00000, 01101) = 3$$

$$d(a, 00011) = d(00000, 00011) = 2$$

$$d(a, 10110) = d(00000, 10110) = 3$$

$$d(a, 11000) = d(00000, 11000) = 2$$

By nearest neighbour decoding rule minimum distance is 2 for $C_x = 00011$

$\therefore$ decoded codeword is 00011

$$b = 01111$$

$$d(b, 01101) = d(01111, 01101) = 1$$

$$d(b, 00011) = d(01111, 00011) = 2$$

$$d(b, 10110) = d(01111, 10110) = 3$$

$$d(b, 11000) = d(01111, 11000) = 4$$

By nearest neighbour decoding rule
minimum distance is 1 for $c_b = 0110$
 $\therefore$ decoded codeword is 01101

iii) $c = 10110$
 $d(c, 00001) = d(10110, 01101) = 4$
 $d(c, 00011) = d(10110, 00011) = 4$
 $d(c, 10110) = d(10110, 10110) = 0$
 $d(c, 11000) = d(10110, 11000) = 3$

By nearest neighbour decoding rule
minimum distance is 0 for $c = 10110$
 $\therefore$ decoded codeword is 10110

iv) $d = 10011$
 $d(d, 01101) = d(10011, 01101) = 4$
 $d(d, 00011) = d(10011, 00011) = 1$
 $d(d, 10110) = d(10011, 10110) = 2$
 $d(d, 11000) = d(10011, 11000) = 3$

By nearest neighbour decoding rule
minimum distance is 1 for $c_d = 00011$
 $\therefore$ decoded codeword is 00011

v) $e = 11011$
 $d(e, 01101) = d(11011, 01101) = 3$
 $d(e, 00011) = d(11011, 00011) = 2$
 $d(e, 10110) = d(11011, 10110) = 3$
 $d(e, 11000) = d(11011, 11000) = 2$

By nearest neighbour decoding rule
minimum distance is 2 for $c_e = 00011$
 $\therefore$ decoded codeword is 00011 or 11000

IMLD

a. Construct incomplete maximum likelihood decoding table for $C = \{000, 111\}$

Received word (x)	$d(x, 000)$	$d(x, 111)$	decoded code
000	0	3	000
001	1	2	000
011	2	1	111
111	3	0	111
010	1	2	000
100	1	2	000
110	2	1	111
101	2	1	111

Construct incomplete maximum likelihood decoding table for $C = \{000, 011\}$

Received word (x)	$d(x, 000)$	$d(x, 011)$	decoded code
000	0	2	000
001	1	1	retransmission
010	1	1	retransmission
100	1	3	000
110	2	2	retransmission
101	2	2	retransmission
011	2	0	011
111	3	1	011

Construct the IMLO table for $C = \{101, 111, 011\}$

Received word (x)	$d(x, 101)$	$d(x, 111)$	$d(x, 011)$	Decoded to
000	2	3	2	Retransmission
001	1	2	1	Retransmission
010	2	2	1	011
100	2	2	3	Retransmission
011	2	1	0	011
110	2	1	2	011
101	0	1	2	101
111	1	0	1	111

Distance of code -

For a code C containing at least two words, the minimum distance of C denoted by $d(C)$

which is given as,

$$d(C) = \min \{d(c_1, c_2) \mid c_1, c_2 \in C, c_1 \neq c_2\}$$

If distance of code is d , length of codeword is n & size of codeword is M then it is referred as (n, M, d) Code

Q. If $C = \{00000, 00111, 11111\}$ is a binary code then find $d(C)$.

$$\begin{aligned} \rightarrow d(00000, 00111) &= 3 \\ d(00000, 11111) &= 5 \\ d(00111, 11111) &= 2 \end{aligned}$$

$$d(C) = \min \{3, 5, 2\} = 2$$

It is $(5, 3, 2)$ code.

n = length of code word
means how many bits of codeword in C
 M = size of codeword

2) Find distance of code where $C = \{000000, 001111, 111222\}$

$$\begin{aligned} d(000000, 001111) &= 4 \\ d(000000, 111222) &= 6 \\ d(001111, 111222) &= 5 \end{aligned}$$

$$d(C) = \min \{4, 6, 5\} = 4$$

It is $(6, 3, 4)$ code.

Role of Hamming distance in Error detection & Error correction.

1) u - Error detecting code -

Theorem :-

Statement - Let u be a positive integer. A code C is u - Error detecting if and only if distance $d(C) \geq u+1$. i.e. A code with distance d is exactly d error detecting.

Ex- $C = \{000, 111\}$

$$d(C) = \{000, 111\} = 3$$

So, A code has minimum distance $d = 3$

$$\therefore \text{It detects } u = d - 1$$

$$= 3 - 1 = 2 \text{ errors}$$

Explanation :

Lets, say the sender sends the codeword 000

Case-I 1-bit error

Recieve : 001

It is not in code C

∴ error detected.

Case-II 2 bit errors

Recieve : 011

It is not in code C

∴ error detected.

Case-III 3 bit errors.

Recieve : 111

This is valid codeword but it's not sent by sender.

∴ error not detected.

∴ By using distance we can detect only $d-1$ errors.

2] V-Error Correcting Code

Let V be a positive integer, a code C is V -error correcting if minimum distance decoding is able to correct V or error assuming incomplete decoding rule is used.

Theorem :

A code C is V -error correcting if and only if distance of $d(C) \geq 2V+1$

i.e code with distance d is exactly $\left\lfloor \frac{d-1}{2} \right\rfloor$ error correcting code where,

$\lfloor x \rfloor \Rightarrow$ greatest integer less equal to x

Relationship to Minimum distance.

- A code can correct upto V -errors if $d \geq 2V+1$.
- So, for 1 error correcting we need $d \geq 2(1)+1 = 3$ i.e $d \geq 3$.

- For 2 error correcting we need $d \geq 2(2)+1 = 5$ i.e $d \geq 5$.

Ex-1) If $C = \{000, 111\}$ then how many errors will be corrected.

→ $d(000, 111) = 3$

Errors corrected = $\left\lfloor \frac{d-1}{2} \right\rfloor = \left\lfloor \frac{3-1}{2} \right\rfloor = \lfloor 1 \rfloor = 1$.

2) Analyze how many error detected & apply we correcting rule on it.

$C = \{000, 111\}$

→ $d(000, 111) = 3$

Errors detected = $d-1 = 3-1 = 2$

Errors corrected = $\left\lfloor \frac{d-1}{2} \right\rfloor = \left\lfloor \frac{3-1}{2} \right\rfloor = 1$

Case-I 1 bit Error -

Sent - 000

Recieved - 001

$(000, 001)$
 Hamming distance to 000 = 1
 Hamming distance to 111 = 2 $(111, 001)$
 Apply nearest neighbour rule
 closest match = 000 (since, hamming distance is 1)

∴ correctly decoded

Case-III 2 bit error :

sent - 000

Received - 011

Hamming distance to 000 = 2

Hamming distance to 111 = 1

Applying nearest neighbour rule

closest match = 111 (since, hamming distance is 1)

∴ incorrectly corrected

Significance of Hamming distance

- 1) Error detection.
- 2) Error correction.
- 3) Design of Reliable codes.
- 4) Measuring code strength.
- 5) Safe communication over noisy channels.
- 6) Used in decoding algorithms.

Unit - III Finite Fields.

Field

A Field is a non empty set F together with two binary operations $+$ (addition) & $\cdot$ (multiplication) satisfying following properties.

$$\forall a, b, c \in F$$

1) Closeness :

F is closed under $+$ & $\cdot$.

$$\text{i.e. } a+b \in F \text{ and } a \cdot b \in F$$

2) Commutative law :

$$a+b = b+a$$

$$a \cdot b = b \cdot a$$

3) Associative law :

$$a+(b+c) = (a+b)+c$$

$$a \cdot (b \cdot c) = (a \cdot b) \cdot c$$

4) Distributive law :

$$a \cdot (b+c) = a \cdot b + a \cdot c$$

5) Additive Identity :

$\exists 0 \in F$ such that,

$$a+0 = a$$

Multiplicative Identity :

$\exists 1 \in F$ such that

$$1 \cdot a = a$$

Real Number: $R = \{Z, Q, Q^+\}$

6) Additive Inverse
For any $a \in F$ $\exists -a \in F$ such that
 $a + (-a) = 0$

Multiplicative Inverse
For any $a \neq 0$ $\exists a^{-1} = \frac{1}{a} \in F$ such that
 $a \cdot a^{-1} = 1$

Ex - Z is a field or not check.

1) $Z =$ set of Integers $= \{-2, -1, 0, 1, 2\}$

It is not field

Because for any element $a \in Z$
there does not exist any multiplicative inverse
of the form $\frac{1}{a}$.

2) Q, R, C all are fields.

• Remark

Let $a, b \in F$ be such that,

1) $(-1) \cdot a = a \cdot (-1) = -a$

2) $a \cdot b = 0 \Rightarrow a = 0$ or $b = 0$

Ex - 1) $15 \equiv 3$

Congruence module m -

Let a, b and $m > 1$. a congruent b
module m , $a \equiv b \pmod{m}$ if $m \mid a-b$

Ex - 1) $15 \equiv 3 \pmod{12}$ — Yes

2) $60 \equiv 30 \pmod{30}$ — Yes

3) $a \equiv 0 \pmod{2}$ means a is even.

4) $a \equiv 1 \pmod{2}$ means a is odd.

Field Z_n (modulo n)

1) $Z_2 = \{0, 1\}$ is a field.

Z_2 satisfy closurness and commutative
property.

It has 0 as a additive identity and
1 as a multiplicative identity.

$0^{-1} = 0$ and $1^{-1} = 1$

$\therefore Z_2$ is a field

2) $Z_3 = \{0, 1, 2\}$

Z_3 is field,

since

For Addition

$$1+2=0$$

$$2+1=0$$

i.e 1 has additive inverse is 2

2 has additive inverse is 1

For multiplication

$$1 \cdot 1 = 1$$

$$2 \cdot 2 = 4 = 1 \pmod{3}$$

i.e 1 has multiplicative inverse is 1

2 has multiplicative inverse is 2

Q.1) Check whether $\mathbb{Z}_4$ is field or not.
using addition & multiplication table.

Soln: $\mathbb{Z}_4 = \{0, 1, 2, 3\}$

Table for addition

+	0	1	2	3
0	0	1	2	3
1	1	2	3	0
2	2	3	0	1
3	3	0	1	2

Table for multiplication

•	0	1	2	3
0	0	0	0	0
1	0	1	2	3
2	0	2	0	2
3	0	3	2	1

2 does not have multiplicative inverse
(there is not any element whose multipli-
cation with 2 is 1)
 $\therefore \mathbb{Z}_4$ is not field.

Q. Construct Addition & multiplication table for
 $\mathbb{Z}_5$ & $\mathbb{Z}_6$

→ i) $\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$

Table for Addition

apsara

Date

apsara

Date

+	0	1	2	3	4
0	0	1	2	3	4
1	1	2	3	4	0
2	2	3	4	0	1
3	3	4	0	1	2
4	4	0	1	2	3

Table for Multiplication

•	0	1	2	3	4
0	0	0	0	0	0
1	0	1	2	3	4
2	0	2	4	1	3
3	0	3	1	4	2
4	0	4	3	2	1

Multiplicative & additive inverse for each
element exist
 $\therefore \mathbb{Z}_5$ is a field.

ii) $\mathbb{Z}_6 = \{0, 1, 2, 3, 4, 5\}$

Table for Addition

+	0	1	2	3	4	5
0	0	1	2	3	4	5
1	1	2	3	4	5	0
2	2	3	4	5	0	1
3	3	4	5	0	1	2
4	4	5	0	1	2	3
5	5	0	1	2	3	4

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Table for multiplication

•	0	1	2	3	4	5
0	0	0	0	0	0	0
1	0	1	2	3	4	5
2	0	2	4	0	2	4
3	0	3	0	3	0	3
4	0	4	2	0	4	2
5	0	5	4	3	2	1

∴ 2, 3, 4 does not have multiplicative inverse
 ∴ $\mathbb{Z}_6$ is not a field.

Imp

Theorem

$\mathbb{Z}_m$ is a field if and only if m is prime

Proof: Suppose $\mathbb{Z}_m$ is a field

claim: m is prime

on contrary suppose m is not prime

i.e. m is a composite number.

∴ $\exists a, b \in \mathbb{Z}$, $1 < a, b < m$
 such that $m = a \cdot b$

In $\mathbb{Z}_m$ $m = 0$

∴ $a \cdot b = 0$

⇒ $a = 0$ or $b = 0$

which is contradiction to the fact that
 $1 < a, b < m$

∴ Our assumption is wrong.

∴ m is prime number.

Conversely, suppose m is a prime,
 claim: $\mathbb{Z}_m$ is a field.

To prove $\mathbb{Z}_m$ is a field it is sufficient
 to prove that for any non-zero
 $a \in \mathbb{Z}_m$, $\exists a^{-1} \in \mathbb{Z}_m$
 such that,

$$a \cdot a^{-1} = 1$$

Let, $a \in \mathbb{Z}_m$

$$\Rightarrow (a, m) = 1 \quad ((a, m) = \gcd \text{ of } a \text{ \& } m)$$

$$\Rightarrow \exists x \text{ \& } y \text{ such that}$$

$$ax + my = 1$$

$$\Rightarrow ax \text{ to } = 1 \quad (\because my = 0 \text{ in } \mathbb{Z}_m)$$

$$\Rightarrow ax = 1$$

$$\Rightarrow x = a^{-1}$$

i.e. for any a there is multiplicative
 inverse of a ,

∴ $\mathbb{Z}_m$ is field.

Q. Find the multiplicative inverse of each of
 the following elements.

2, 5, 8 in $\mathbb{Z}_{11}$

$$\rightarrow \mathbb{Z}_{11} = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$$

$$2 \cdot 1 = 2$$

$$2 \cdot 2 = 4$$

$$2 \cdot 3 = 6$$

$$2 \cdot 4 = 8$$

$$2 \cdot 5 = 10$$

$$2 \cdot 6 = 12 \equiv 1 \pmod{11} \quad \text{i.e. } 2^{-1} = 6$$

for 5

$$5 \cdot 1 = 5$$

$$5 \cdot 2 = 10$$

$$5 \cdot 3 = 15$$

$$5 \cdot 4 = 20$$

$$5 \cdot 5 = 25$$

$$5 \cdot 6 = 30$$

$$5 \cdot 7 = 35$$

$$5 \cdot 8 = 40$$

$$5 \cdot 9 = 45 \equiv 1 \pmod{11} \quad \text{i.e. } 5^{-1} = 9$$

for 8

$$8 \cdot 7 = 56 \equiv 1 \pmod{11} \quad \text{i.e. } 8^{-1} = 7$$

Q. 2, 3, 4 In $\mathbb{Z}_7$
 $\mathbb{Z}_7 = \{0, 1, 2, 3, 4, 5, 6\}$

For 2

$$2 \cdot 4 = 8 \equiv 1 \pmod{7} \quad \text{i.e. } 2^{-1} = 4$$

For 3

$$3 \cdot 5 = 15 \equiv 1 \pmod{7} \quad \text{i.e. } 3^{-1} = 5$$

For 4

$$4 \cdot 2 = 8 \equiv 1 \pmod{7} \quad \text{i.e. } 4^{-1} = 2$$

Ass. 2

- a. 1) Construct addition & multiplication tables for $\mathbb{Z}_8$ & verify it is a field or not.
2) Find multiplicative inverse of 4, 7 & 11 in $\mathbb{Z}_{17}$.

Characteristics of Field

- Let F be a field, the characteristic of field F is least positive integer p such that $p \cdot 1 = 0$, where 1 is a multiplicative identity of F .
- If no such p exist then it is equal to 0.

Ex-1) $\text{char } \mathbb{A} = 0$
 $\text{char } \mathbb{R} = 0$
 $\text{char } \mathbb{C} = 0$

Since, there does not exist any least positive integer p such that $p \cdot 1 = 0$.

2) $\text{char } \mathbb{Z}_2 = 2$
Since $2 \cdot 1 = 2 \equiv 0$ in $\mathbb{Z}_2$

3) $\text{char } \mathbb{Z}_3 = 3$
Since $3 \cdot 1 = 3 \equiv 0$ in $\mathbb{Z}_3$

$\therefore \text{char of } \mathbb{Z}_n = n$ (n is prime)

Theorem

Characteristics of Field is either 0 or a prime number.

Proof: since $1 \cdot 1 = 1 \neq 0$ therefore 1 is not characteristic of any field.

Suppose p is characteristic of field.

claim: p is prime number.
 one contrary assume that p is a composite number.

$\therefore p = n \cdot m$, $1 \leq n, m < p$
 where n & m are primes.
 since, p is a characteristic of field
 $\therefore p \cdot 1 = 0$
 $\Rightarrow (n \cdot m) \cdot 1 = 0$
 $\Rightarrow (n \cdot 1) \cdot (m \cdot 1) = 0$
 $\Rightarrow n \cdot 1 = 0$ or $m \cdot 1 = 0$
 $\Rightarrow n$ or m is characteristic of field. But
 $1 \leq n, m < p$ & p is least positive integer.
 since, by definition of characteristic
 $\therefore$ We get contradiction.
 $\therefore p$ is a prime number.

* Ring -

The set F which follows properties from 1 to 6 of field are called rings.

Ex - $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, $\mathbb{C}$ all are rings.

* Polynomial rings :-

Let F be a field, The set
 $F[x] = \left\{ \sum_{i=0}^n a_i x^i \mid a_i \in F, n \geq 0 \right\}$

is called the polynomial ring over F .

An element of $f(x)$ is called polynomial over F .
 The general form of polynomial is,

$$f(x) = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

for a polynomial $f(x) = \sum_{i=0}^n a_i x^i$

$= a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$
 the integer n is called the degree of $f(x)$,
 denoted by $\deg(f(x))$.

Ex - 1) $5x^3 + 4x^6 + 7x^7 + 2x^8$ has degree 8.
 2) $x^2 + 1$ has degree 2.

* Monic Polynomial

A non-zero polynomial

$$f(x) = \sum_{i=0}^n a_i x^i = a_0 + a_1 x + a_2 x^2 + \dots + a_n x^n$$

is said to be monic if $a_n = 1$ of degree

Ex - i) $f(x) = 5x^3 + 4x^6 + 7x^7 + 2x^8$ is not monic
 Since, $a_8 = 2 \neq 1$

ii) $f(x) = x^2 + 1$ is a monic polynomial since,
 $a_2 = 1$

* Reducible Polynomial

A polynomial $f(x)$ of positive degree is said to be reducible over F if there exist two polynomials $g(x)$ & $h(x)$.

such that,
 $f(x) = g(x) \cdot h(x)$ where
 $1 \leq \deg(g(x))$,
 $\deg(h(x)) < \deg(f(x))$

otherwise the polynomial $f(x)$ of positive degree is said to be irreducible polynomial over F .

Ex- i) $f(x) = x^2 + x \in \mathbb{R}[x]$ is a reducible polynomial because,
 $f(x) = x^2 + x = x(x+1)$

ii) $f(x) = x^2 - 2 \in \mathbb{Q}[x]$ is an irreducible polynomial over $\mathbb{Q}$, because
 $f(x) = x^2 - 2 = (x + \sqrt{2})(x - \sqrt{2})$
 but $\pm \sqrt{2} \notin \mathbb{Q}$

iii) $f(x) = x^2 - 2 \in \mathbb{R}[x]$ is a reducible polynomial because,
 $f(x) = x^2 - 2 = (x + \sqrt{2})(x - \sqrt{2})$,
 $\sqrt{2} \pm \sqrt{2} \in \mathbb{R}$

iv) $f(x) = x^2 + 1 \in \mathbb{R}[x]$ is irreducible polynomial, because
 $f(x) = x^2 + 1$
 $x^2 + 1 = 0$, $x^2 = -1 = i^2$
 $x = \pm i$ but $i \notin \mathbb{R}$

v) $f(x) = x^2 + 1 \in \mathbb{C}[x]$ is reducible polynomial, because

$$x^2 + 1 = 0, \quad x^2 = -1 = i^2$$

$$x = \pm i, \quad i \in \mathbb{C}$$

Remark :-

The polynomial $f(x)$ is an irreducible polynomial in a field $\mathbb{F}$ if it has no roots in that specific field. otherwise polynomial is reducible.

Ex- Check whether the polynomial $f(x) = x^4 + 2x^2$ is reducible or irreducible over $\mathbb{Z}_3[x]$.

$\rightarrow \mathbb{Z}_3 = \{0, 1, 2\}$
 $f(0) = (0)^4 + 2(0)^2 = 0$
 $f(1) = 1^4 + 2(1)^2 = 1 + 2 = 3 = 0$ in $\mathbb{Z}_3$
 $f(2) = 2^4 + 2(2)^2 = 16 + 8 = 24 = 0$ in $\mathbb{Z}_3$
 $\therefore 0, 1, 2$ are roots of $f(x)$
 $\therefore f(x)$ is reducible polynomial.

2) Check whether the polynomial $f(x) = 1 + x + x^2 \in \mathbb{Z}_2[x]$ is reducible or irreducible polynomial.

$\rightarrow \mathbb{Z}_2 = \{0, 1\}$
 $f(0) = 1 + 0 + 0 = 1$ in $\mathbb{Z}_2$
 $f(1) = 1 + 1 + 1 = 3 = 1$ in $\mathbb{Z}_2$
 $\therefore 0, 1$ are not roots of $f(x)$
 $\therefore f(x)$ is irreducible polynomial.

3) Check whether the polynomial $f(x) = 1 + x^2 + x^3 \in \mathbb{Z}_3[x]$ is reducible or irreducible.

$\rightarrow \mathbb{Z}_3 = \{0, 1, 2\}$

$$\begin{aligned}
 f(0) &= 1+0^2+0^3 = 1 \text{ in } \mathbb{Z}_3 \\
 f(1) &= 1+1^2+1^3 = 3 = 0 \text{ in } \mathbb{Z}_3 \\
 f(2) &= 1+2^2+2^3 = 13 = 1 \text{ in } \mathbb{Z}_3
 \end{aligned}$$

$\therefore 0, 2$ are not roots of $f(x)$
 $\therefore f(x)$ is irreducible polynomial.

Defⁿ :-
 Let $f(x) \in F[x]$ be a polynomial of degree $n \geq 1$.
 Then for any polynomial $g(x) \in F[x]$,
 there exist unique pair $s(x), r(x)$ of
 polynomials with $\deg r(x) < \deg f(x)$ or
 $r(x) = 0$ such that $g(x) = s(x)f(x) + r(x)$.

Then polynomial $r(x)$ is called the remainder
 of $g(x)$ divided by $f(x)$.

Defⁿ
 Let $f(x), g(x) \in F[x]$ be two non-zero
 polynomials. the greatest common divisor
 of $f(x), g(x)$ is denoted by $\gcd$
 $(f(x), g(x))$, is the monic polynomial
 of the highest degree which is a divisor
 of both $f(x)$ & $g(x)$.
 In particular we say that $f(x)$ is co-prime
 to $g(x)$ if $\gcd(f(x), g(x)) = 1$.

Imp Q. Construct addition & multiplication table
 for $\mathbb{Z}_2[x] / 1+x^2$

$\rightarrow$ since $1+x^2$ has degree 2.
 $\therefore \mathbb{Z}_2[x] / 1+x^2$ has elements of the form
 $f(x) = a_0 + a_1x$
 $\mathbb{Z}_2 = \{0, 1\}$
 $a_0 = a_1 = 0 \Rightarrow f(x) = a_0 + a_1x = 0$
 $a_0 = 0, a_1 = 1 \Rightarrow f(x) = a_0 + a_1x = x$
 $a_0 = 1, a_1 = 1 \Rightarrow f(x) = a_0 + a_1x = 1+x$
 $a_0 = 1, a_1 = 0 \Rightarrow f(x) = a_0 + a_1x = 1$

$$\frac{\mathbb{Z}_2[x]}{1+x^2} = \{0, 1, x, 1+x\}$$

for addition

+	0	1	x	1+x
0	0	1	x	1+x
1	1	0	1+x	x
x	x	1+x	0	1
1+x	1+x	x	1	0

For Multiplication

x	0	1	x	1+x
0	0	0	0	0
1	0	1	x	1+x
x	0	x	1	1+x
1+x	0	1+x	1+x	0

Given polynomial is $1+x^2$

$$1+x^2 = 0 \Rightarrow x^2 = -1,$$

-1 is not exist in $\mathbb{Z}_2[x] / 1+x^2$.

we know that,

$a \equiv b \pmod{m} \Rightarrow m \mid a-b$
 We have to find, $-1 = ? \pmod{2}$
 $\mathbb{Z}_2 = \{0, 1\}$
 $-1 \not\equiv 0 \pmod{2}$
 $-1 \equiv 1 \pmod{2}$
 $\therefore x^2 = 1$ in $\mathbb{Z}_2$

$(\mathbb{Z}_2[x] / 1+x^2)$ is not field because $1+x$ has not multiplicative inverse.

Q. Construct addition and multiplication table for $\mathbb{Z}_2[x] / 1+x+x^2$
 $\rightarrow$ We know that $\mathbb{Z}_2[x] / 1+x+x^2 = \{0, 1, x, 1+x\}$

Addition table is

+	0	1	x	1+x
0	0	1	x	1+x
1	1	0	1+x	x
x	x	1+x	0	1
1+x	1+x	x	1	0

Multiplication table is :-

•	0	1	x	1+x
0	0	0	0	0
1	0	1	x	1+x
x	0	x	1+x	1
1+x	0	1+x	1	x

$$\begin{aligned}
 x(1+x) &= x+x^2 \\
 x+1 &= x \\
 1+x+x^2 &= 0 \\
 x^2 &= 1-x \\
 x^2 &= 1+x \\
 [x+x^2 = -1] \text{ in } \mathbb{Z}_2
 \end{aligned}$$

$\mathbb{Z}_2[x] / 1+x+x^2$ is a field because all the elements has multiplicative inverse.

$$\begin{aligned}
 \text{i)} \quad 1+x+x^2 &= 0 \\
 x^2 &= -1-x \\
 x^2 &= 1+x \text{ in } \mathbb{Z}_2 \\
 (-1 &\equiv 1 \pmod{2}) \\
 \text{ii)} \quad 1+x+x^2 &= 0 \\
 x+x^2 &= -1 \equiv 1 \text{ in } \mathbb{Z}_2 \\
 (1+x)(1+x) &= 1+2x+x^2 \text{ or } 1+2x+x^2 = x^2+1 \\
 1+0+1 &= x \\
 &= 2+x \\
 &= 0+x \\
 &= x
 \end{aligned}$$

Structure of Finite Fields

Finite field is denoted by F_p^n
 Where $F_p^n = F_p[x] / f(x)$
 n is degree of the polynomial.

Ex- $F_2^2 = F_4 = F_2[x] / 1+x+x^2 = \{0, 1, x, 1+x\}$ is a finite field.

Remark :- For every element β of finite field with q elements we have $\beta^q = \beta$

Ex-1) $F_4 = F_2[x] / 1+x+x^2 = \{0, 1, x, 1+x\}$ is a finite field.

because F_4 has 4 elements

$$\therefore q = 4$$

check condition $\beta^q = \beta$

$$\therefore 0^4 = 0$$

$$1^4 = 1$$

$$x^4 = ?$$

$$1+x+x^2=0$$

$$x^2 = -1-x = 1+x \text{ in } \mathbb{Z}_2$$

$$x^4 = x^2 \cdot x^2$$

$$= (1+x) \cdot (1+x)$$

$$= 1+2x+x^2$$

$$= 1+x^2$$

$$= 1+1+x$$

$$= 2+x$$

$$= x$$

$$(1+x)^4 = (1+x^2) \cdot (1+x)^2$$

$$= (1+2x+x^2) (1+2x+x^2)$$

$$= (1+x)^2 (1+x)^2 \text{ by multiply}$$

$$= 1+x^2+x^2+x^4$$

$$= 1+2x^2+x^4$$

$$(1+x)^4 = 1+x$$

definition : An element α in a finite field F_q is called generator of F_q if $F_q = \{0, \alpha, \alpha^2, \dots, \alpha^{q-1}\}$

Ex - We have $F_4 = \{0, 1, x, 1+x\}$
 x is generator of F_4
 because, F_4 has 4 elements

$$\therefore q = 4$$

$$x^1 = x$$

$$x^2 = 1+x$$

$$x^3 = x \cdot x^2 = x \cdot (1+x) = x+x^2$$

$$= x+1+x$$

$$= 2x+1$$

$$\therefore x^2 = 1+x$$

$$\therefore x^2 = 0 \text{ in } \mathbb{Z}_2$$

$$\mathbb{Z}_2$$

$$\therefore F_4 = \{0, x, x^2, x^3\}$$

$$= \{0, x, 1+x, 1\}$$